Answers and talk notes

Apply Two-Variable Relationships to Solve Problems

What your student is learning

Tonight we start from the GOAL and work backward. Put the target in place of the output, then undo the operations one at a time to find the input that reaches it.

Answers

Ask your student to explain before you look.

1. **A.** 26n = 130. 26n = 130, because the goal goes where the output was.
2. **B.** 5. n = 5, because 130 ÷ 26 = 5.
3. **C.** Dividing by 9. Dividing by 9 undoes multiplying by 9.
4. **D.** 12w = 96. 12w = 96, because the goal replaces s.
5. **B.** 8. w = 8, because 96 ÷ 12 = 8.
6. **C.** 96. s = 96, because 12 × 8 = 96, which matches the goal.

Talk about it

What is your goal, and where did you put it in the equation? Which operation undid the multiplication?

Home mission

Pretend you save $4 each week toward a $48 goal. Can your student write 4w = 48 and undo the multiplying? Then put the answer back: do that many weeks of $4 really make $48?

When they are stuck, ask

- What is the goal, and where does it go in the equation?
- What was done to the variable? What undoes that?
- Put your answer back in. Does it reach the goal?

Try not to say

- Just multiply the two numbers in the problem.
- Guess a number and see if it works — keep guessing.
- The bigger number always gets divided by the smaller one.